Group Project: Airlines Management System

User Guide

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1. System Overview

The SkyHigh Airlines Management System is a console-based Java application that simulates core airline operations. It follows a clean object-oriented design based on the **Stage 3 UML diagram** (in GitHub Repo).

Key design principles used:

- **Single Responsibility Principle (SRP)**: Each class has one clear job (entities hold data, managers handle collections, MainTester manages UI and testing).
- **Management Classes**: AirportManager, StaffManager, BookingManager, and TicketManager coordinate collections and interactions.
- **CRC Card Approach**: Responsibilities and collaborators were clearly defined during development.
- **Light Coordinator Pattern**: Airline class only holds references to managers.

The system starts with dummy data creation (industry-standard happy path), displays a system report, and then enters an interactive menu for navigation.

2. How to Run the System

1. Ensure all project classes (Airline.java, Airport.java, Staff.java, Flight.java, etc.) are in the same folder.
2. Compile and run MainTester.java.
3. The program automatically:
    - Creates initial dummy objects (passengers, flights, staff, bookings, etc.).
    - Displays a consolidated System Report.
    - Opens the main menu.

Important Workflow: MainTester acts as the single entry point. It calls runHappyPathTest() to build objects, then displaySystemReport(), and finally runInteractiveMenu(). All business logic stays inside the dedicated manager and entity classes.

[Screenshot 1: Startup Happy-Path Output + System Report]

```java
    /**
     * Runs the complete happy-path integration test
     * Called once at startup and again via menu option 6
     */
    private static void runHappyPathTest() {
        // 1. Lite Coordinator (Airline)
        airline = new Airline("SkyHigh Airlines");
        AirportManager airportManager = airline.getAirportManager();
        StaffManager staffManager = airline.getStaffManager();
        BookingManager bookingManager = airline.getBookingManager();
        TicketManager ticketManager = airline.getTicketManager();

        // 2. LABEEB - Infrastructure
        Airport phx = new Airport("PHX", "Phoenix Sky Harbor", "Phoenix, AZ, USA");
        Airport lax = new Airport("LAX", "Los Angeles International", "Los Angeles, CA, USA");
        airportManager.addAirport(phx);
        airportManager.addAirport(lax);

        Terminal t4 = new Terminal("T4", "Terminal 4");
        Terminal t5 = new Terminal("T5", "Terminal 5");

        Aircraft a320 = new Aircraft("A320-001", "Airbus A320", 180);
        a320.generateSeats();          // NEW
        Aircraft b737 = new Aircraft("B737-002", "Boeing 737", 160);
        b737.generateSeats();          // NEW

        // 3. RUIZ - Schedule & Flight
        Schedule schDep = new Schedule("SCH-001", "2026-04-15 08:00", "2026-04-15 10:30");
        Schedule schArr = new Schedule("SCH-002", "2026-04-15 11:00", "2026-04-15 13:30");

        flightAA123 = new Flight("AA123", phx, lax, t4, t5, schDep, schArr, "ON_TIME", a320);

        // 3b. Create FlightAssignment (HEIFNER + RUIZ integration)
        FlightAssignment fa = new FlightAssignment("FL-ASSIGN-AA123", flightAA123, a320);
        flightAA123.setFlightAssignment(fa);   // Links back to the Flight

        System.out.println("FlightAssignment created and linked: " + fa.getAssignmentID());

        // 4. HEIFNER - Personnel
        captain = new Staff("Captain Elena Ruiz", "elena.ruiz@skyhigh.com", "S-001", "Pilot");
        attendant = new Staff("Maria Lopez", "maria.lopez@skyhigh.com", "S-002", "Flight Attendant");
```

3. Main Menu

After startup, the interactive menu appears. Use number keys to navigate. Type 0 to quit.

[Screenshot 2: Main Menu]

```
=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
Enter choice: Enter choice (0-6):
```

4. Important Sections

4.1 Infrastructure Management (Airports, Terminals, Aircraft)

- View airports and aircraft details.
- Check available seats on the current flight.

[Screenshot 3: Infrastructure Submenu]

```
--- INFRASTRUCTURE MANAGEMENT ---
1. List All Airports
2. View Aircraft Details
3. Show Available Seats on Current Flight
0. Back to Main Menu
Enter choice (0-3): 1
Airports in system:
 - PHX: Phoenix Sky Harbor
 - LAX: Los Angeles International

--- INFRASTRUCTURE MANAGEMENT ---
1. List All Airports
2. View Aircraft Details
3. Show Available Seats on Current Flight
0. Back to Main Menu
Enter choice (0-3): 2
Aircraft Inventory:
 - A320-001 | Airbus A320 | Capacity 180
 - B737-002 | Boeing 737 | Capacity 160

--- INFRASTRUCTURE MANAGEMENT ---
1. List All Airports
2. View Aircraft Details
3. Show Available Seats on Current Flight
0. Back to Main Menu
Enter choice (0-3): 3
Available seats on AA123: 178

--- INFRASTRUCTURE MANAGEMENT ---
1. List All Airports
2. View Aircraft Details
3. Show Available Seats on Current Flight
0. Back to Main Menu
```

4.2 Personnel & Assignments (Staff & Passengers)

- View staff schedules.
- List passengers.
- Show crew assigned to Flight AA123.

[Screenshot 4: Personnel Submenu]

```
=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
Enter choice: Enter choice (0-6): 2

--- PERSONNEL & ASSIGNMENTS ---
1. View Staff Schedule
2. List All Passengers
3. Show Crew for Flight AA123
0. Back to Main Menu
Enter choice (0-3): 1
Captain Elena Ruiz Schedule:
Schedule SCHEDULE-S-001: DEPARTURE TIME ? ARRIVAL TIME

Maria Lopez Schedule:
Schedule SCHEDULE-S-002: DEPARTURE TIME ? ARRIVAL TIME

--- PERSONNEL & ASSIGNMENTS ---
1. View Staff Schedule
2. List All Passengers
3. Show Crew for Flight AA123
0. Back to Main Menu
Enter choice (0-3): 2
Passengers in system:
 - P-1001: John Heffner
 - P-1002: Eduardo Ceh-Varela
```

```
--- PERSONNEL & ASSIGNMENTS ---
1. View Staff Schedule
2. List All Passengers
3. Show Crew for Flight AA123
0. Back to Main Menu
Enter choice (0-3): 3
Crew assigned to AA123: 2 members
 - Captain Elena Ruiz (Pilot)
 - Maria Lopez (Flight Attendant)

--- PERSONNEL & ASSIGNMENTS ---
1. View Staff Schedule
2. List All Passengers
3. Show Crew for Flight AA123
0. Back to Main Menu
Enter choice (0-3): 0
```

4.3 Flight Operations

- View current flight details (AA123).
- Check flight status and aircraft assignment.

[Screenshot 5: Flight Operations Submenu]

```
=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
Enter choice: Enter choice (0-6): 3

--- FLIGHT OPERATIONS ---
1. View Current Flight Details
2. Show Flight Status
3. Check Aircraft Assignment
0. Back to Main Menu
Enter choice (0-3): 1
Flight: AA123 | From: PHX | To: LAX | Status: ON_TIME | Assignment: FL-ASSIGN-AA123

--- FLIGHT OPERATIONS ---
1. View Current Flight Details
2. Show Flight Status
3. Check Aircraft Assignment
0. Back to Main Menu
Enter choice (0-3): 2
Flight AA123 Status: ON_TIME

--- FLIGHT OPERATIONS ---
1. View Current Flight Details
2. Show Flight Status
3. Check Aircraft Assignment
0. Back to Main Menu
Enter choice (0-3): 3
Aircraft assigned to AA123: A320-001 (Airbus A320)

--- FLIGHT OPERATIONS ---
1. View Current Flight Details
2. Show Flight Status
3. Check Aircraft Assignment
0. Back to Main Menu
Enter choice (0-3): 0

=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
```

4.4 Bookings & Tickets

- Shows **all booking and ticket status** for passengers (John Heffner and Eduardo Ceh-Varela).
- Displays real ticket details pulled from TicketManager and Booking objects.

[Screenshot 6: Bookings & Tickets Submenu]

```
=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
Enter choice: Enter choice (0-6): 4

--- BOOKINGS & TICKETS ---
1. Show All Booking Status for Tickets
0. Back to Main Menu
Enter choice (0-1): 1

=== ALL BOOKINGS & TICKETS STATUS ===
Ticket ID: T1776049939917 | Flight: AA123 | Seat: 1 | Status: CONFIRMED | Passenger: John Heffner
Ticket ID: T1776049939917 | Flight: AA123 | Seat: 2 | Status: CONFIRMED | Passenger: Eduardo Ceh-Varela

Total Tickets: 2
Booking ID: B1776049939917 | Status: CONFIRMED
=== END OF BOOKING/TICKET STATUS ===

--- BOOKINGS & TICKETS ---
1. Show All Booking Status for Tickets
0. Back to Main Menu
Enter choice (0-1): 0

=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
```

4.5 Payments & Reports

- Re-display the System Report.
- View payment details for the current booking.

[Screenshot 7: Payments & Reports Submenu]

```
=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
Enter choice: Enter choice (0-6): 5

--- PAYMENTS & REPORTS ---
1. Show System Report (Summary)
2. View Payment Details
0. Back to Main Menu
Enter choice (0-2): 1

=== SKYHIGH AIRLINES SYSTEM REPORT ===
Customers/Passengers created: John Heffner, Eduardo Ceh-Varela
Transactions/Payments processed: $449.98 total revenue
Active Flights: AA123 (ON_TIME)
Available Seats on AA123: 178
Staff Assigned: Captain Elena Ruiz (Pilot), Maria Lopez (Attendant)
Bookings Created: B1776049939917 (CONFIRMED)
=== REPORT COMPLETE ===


--- PAYMENTS & REPORTS ---
1. Show System Report (Summary)
2. View Payment Details
0. Back to Main Menu
Enter choice (0-2): 2
Payment for B1776049939917 | Amount: $299.99 | Status: PENDING
Payment for B1776049939917 | Amount: $149.99 | Status: PENDING

--- PAYMENTS & REPORTS ---
1. Show System Report (Summary)
2. View Payment Details
0. Back to Main Menu
Enter choice (0-2): 0
```

4.6 Re-run Full Happy-Path Integration Test

- Re-displays the complete original test output (all numbered sections, passenger tickets, staff schedules, flight details, payments, etc.).

[Screenshot 8: Re-run Test Output]

```
=== MAIN MENU (SkyHigh Airlines) ===
1. Infrastructure Management (Airports / Terminals / Aircraft)
2. Personnel & Assignments (Staff / Passengers)
3. Flight Operations
4. Bookings & Tickets
5. Payments & Reports
6. Re-run Full Happy-Path Integration Test (original test output)
0. Quit
Enter choice: Enter choice (0-6): 6

=== RE-RUNNING FULL HAPPY PATH INTEGRATION TEST ===
FlightAssignment created and linked: FL-ASSIGN-AA123

--- STAFF VIEW SCHEDULE (HEIFNER CRC) ---
Captain Elena Ruiz Schedule:
Schedule SCHEDULE-S-001: DEPARTURE TIME ? ARRIVAL TIME

Maria Lopez Schedule:
Schedule SCHEDULE-S-002: DEPARTURE TIME ? ARRIVAL TIME

--- PASSENGER VIEW TICKETS (HEIFNER CRC) ---
Ticket ID: T1776050302517 | Flight: AA123 | Seat: 1 | Status: CONFIRMED | Passenger: John Heffner
Ticket ID: T1776050302517 | Flight: AA123 | Seat: 2 | Status: CONFIRMED | Passenger: Eduardo Ceh-Varela

--- STAFF VIEW SCHEDULE (HEIFNER CRC) ---
Staff assignments for Captain: 2 crew member(s)

--- FLIGHT DETAILS (RUIZ CRC) ---
Flight: AA123 | From: PHX | To: LAX | Status: ON_TIME | Assignment: FL-ASSIGN-AA123
Available seats: 178

--- PAYMENT DETAILS ---
Payment ID: PAY-1776050302518 | Amount: $299.99 | Status: PENDING | Booking: B1776050302517
Payment ID: PAY-1776050302518 | Amount: $149.99 | Status: PENDING | Booking: B1776050302517

=== TEST COMPLETE. All Dummy objects created and linked ===
```

5. System Design Summary

- **UML Foundation**: The class diagram shows clear relationships (Airline owns managers; Flight links to Schedule, Aircraft, and StaffAssignment; Booking contains Tickets, etc.).
- **Class Responsibilities**:
  - **Entities** (Flight, Passenger, Staff, Ticket, etc.): Hold data and basic behaviors.
  - **Managers**: Handle collections and coordination.
  - **MainTester**: Only handles startup, reporting, and user navigation.

- This design ensures modularity and maintainability.

5.1 Design Principles Demonstrated

- The SkyHigh Airlines Management System demonstrates key object-oriented principles. **Inheritance** is clearly shown through the **Person** abstract class, which provides shared attributes (**name** and **contactInfo**) and behavior (**updateContactInfo()**) to both **Staff** and **Passenger** subclasses. This avoids code duplication while allowing each subclass to add specialized fields and methods (e.g., **staffID/role** for **Staff** and **passengerID/passportNumber** for **Passenger**).
- **Encapsulation** is applied throughout: all instance variables are private, accessed only via public getter/setter methods, and managers return defensive copies of collections (e.g., **TicketManager.getAllTickets()**). This protects internal data and maintains clean boundaries between classes.

6. Exit the System

Select **0** from the Main Menu to quit.

Thank you for using SkyHigh Airlines Management System!